UM-SJTU Joint Institute

Electronic Circuits

(ECE3110J)

July 12, 2024

Laboratory Report

Lab 3

Group Members:

Jiacheng Yin

Jiazhang Ping

Zifeng Li

Contents

1 Objectives

The purposes of this lab are as follows:

- Develop better understanding towards MOSFET circuits.
- Learn about the source degeneration and its effects.
- Practice the design of a common-source amplifier.

2 Experimental Results

2.1 I-V Characteristics of 2N7000

The circuit to measure the early effect voltage is as follows:

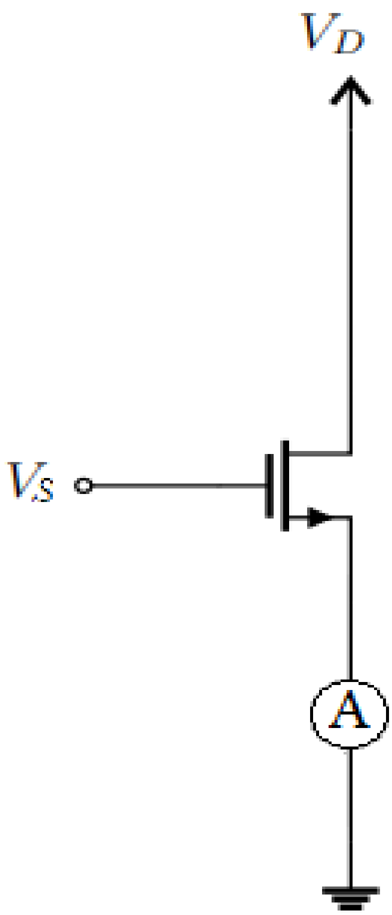

Figure 1: I-V Characteristic Testing Circuit

V_G =1.0V	$V_D(V)$	1.5	2.0	2.5	3.0	3.5	4.0
	$I(mA)$	2.58	16.58	18	18.27	18.44	18.50
V_G =1.2V	$V_D(V)$	1.5	2.0	2.5	3.0	3.5	4.0
	$I(mA)$	3.15	19.74	21.58	21.98	22.12	22.2

Figure 2: Measured Currents with Fixed V_G

V_D =1.5V	$V_G(V)$	0.8	0.9	1.0	1.1	1.2	0.8
	$I(mA)$	0.47	3.00	10.74	27.32	55.2	0.47
V_D =2V	$V_G(V)$	0.8	0.9	1.0	1.1	1.2	0.8
	$I(mA)$	0.55	3.15	11.23	28.96	59.7	0.55

Figure 3: Measured Currents with Fixed V_D

The corresponding I-V curves are as follows:

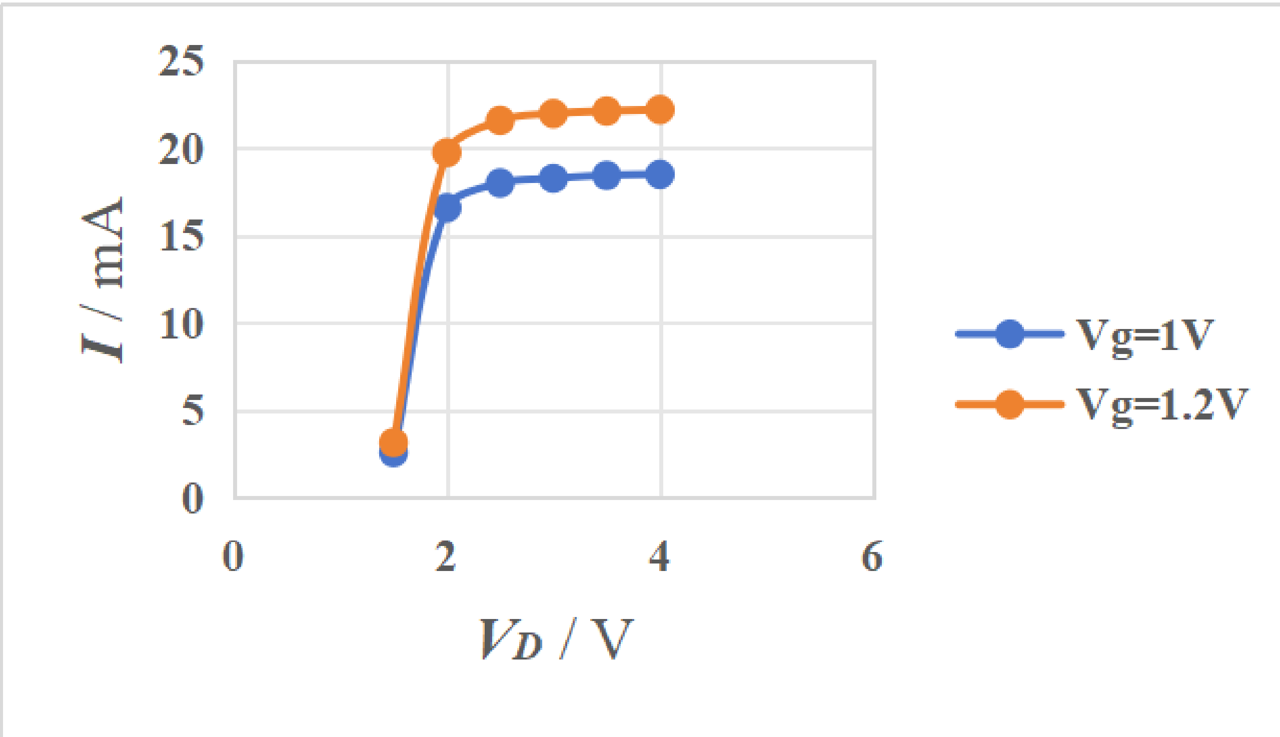

Figure 4: Experimental I-V Curve of 2N7000 (1)

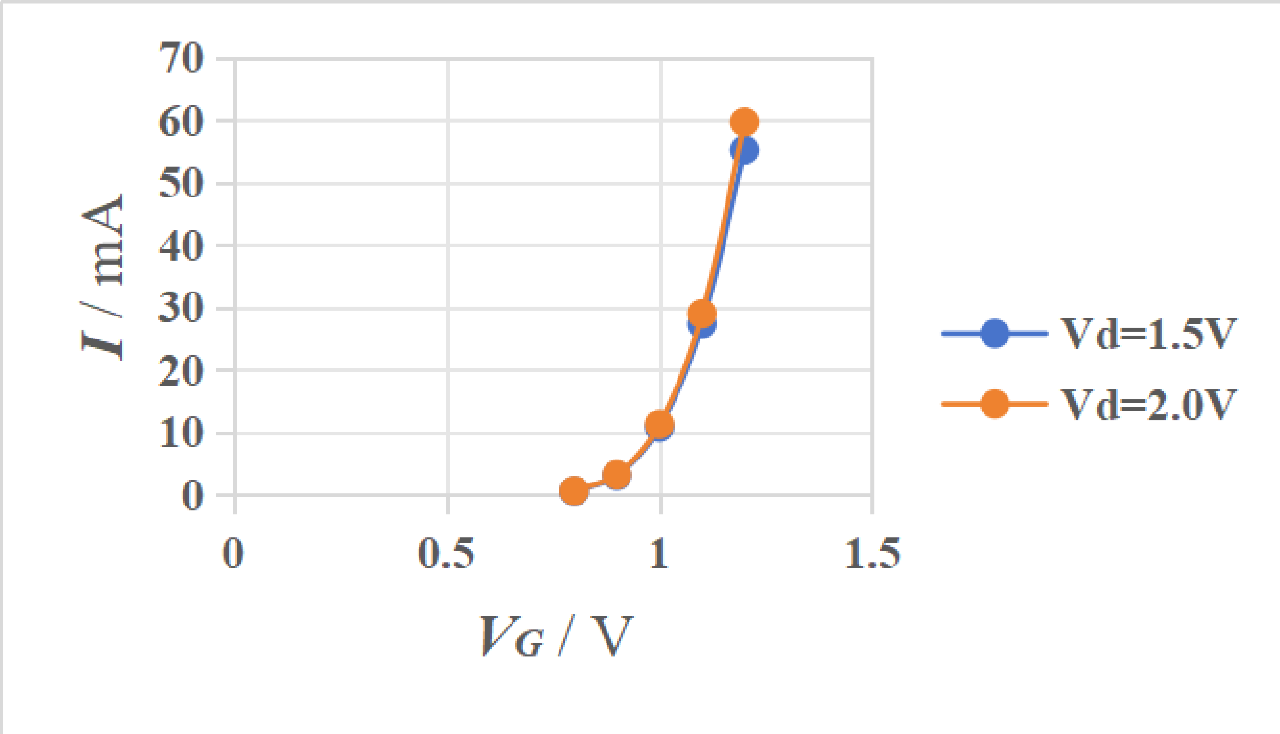

Figure 5: Experimental I-V Curve of 2N7000 (2)

2.2 Common-Source Amplifier with Source Degeneration

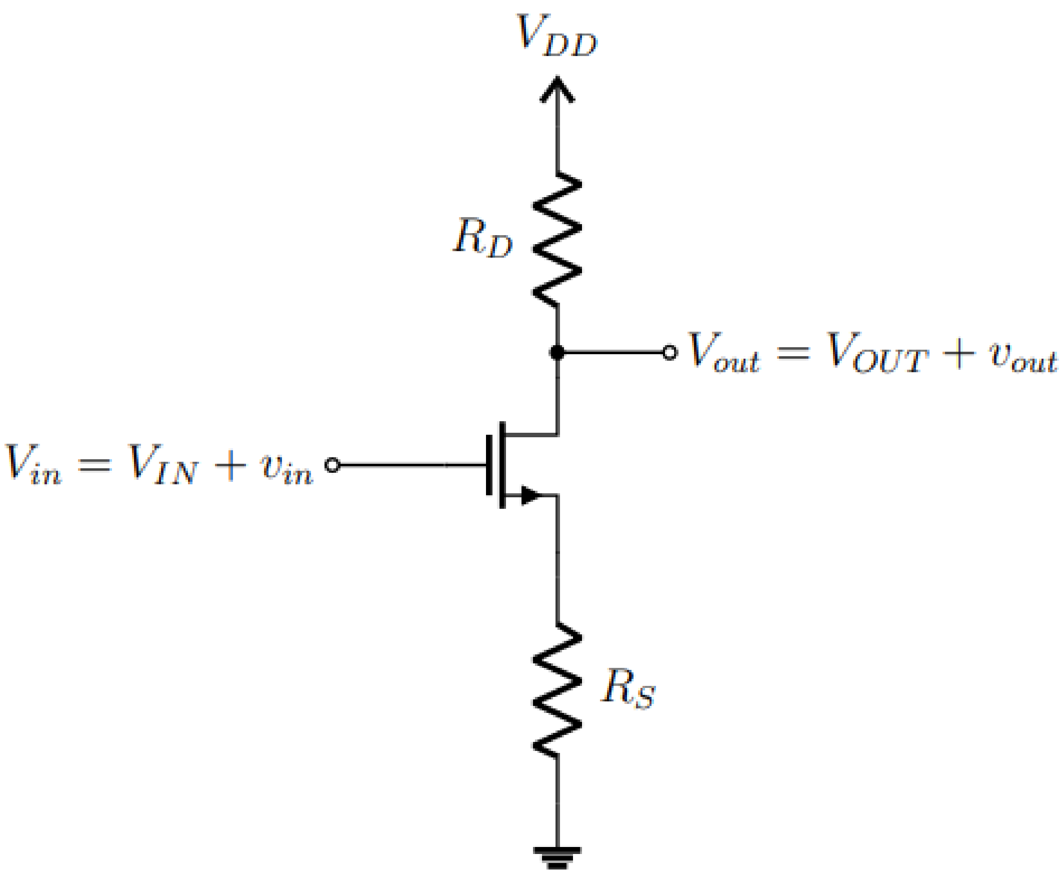

Figure 6: The Common-Source Amplifier Circuit

Select Values:

I	V_{DD}	V_{IN}	R_D	R_S
1 mA	10 V	2 V	7500 Ω	1200 Ω

Using the relationships:

$$\begin{aligned} V_D &= V_{DD} - I_D R_D \\ &= 10\text{ V} - (0.001\text{ A} \times 7500\Omega) \\ &= 10\text{ V} - 7.5\text{ V} \\ &= 2.5\text{ V} \end{aligned}$$

$$\begin{aligned} V_S &= I_D R_S \\ &= 0.001\text{ A} \times 1200\Omega \\ &= 1.2\text{ V} \end{aligned}$$

$$\begin{aligned} V_{DS} &= V_D - V_S \\ &= 2.5\text{ V} - 1.2\text{ V} \\ &= 1.3\text{ V} \end{aligned}$$

$$\begin{aligned} V_{GS} &= V_{IN} - V_S \\ &= 2\text{ V} - 1.2\text{ V} \\ &= 0.8\text{ V} \end{aligned}$$

	V_D	V_S	V_{DS}	V_{GS}
Theoretical	2.5 V	1.2 V	1.3 V	0.8 V
Actual	6.9 V	0.494 V	6.41 V	1.513 V

Finally, we apply a small AC voltage on the input. The results are shown as follows:

Figure 7: Vin 1

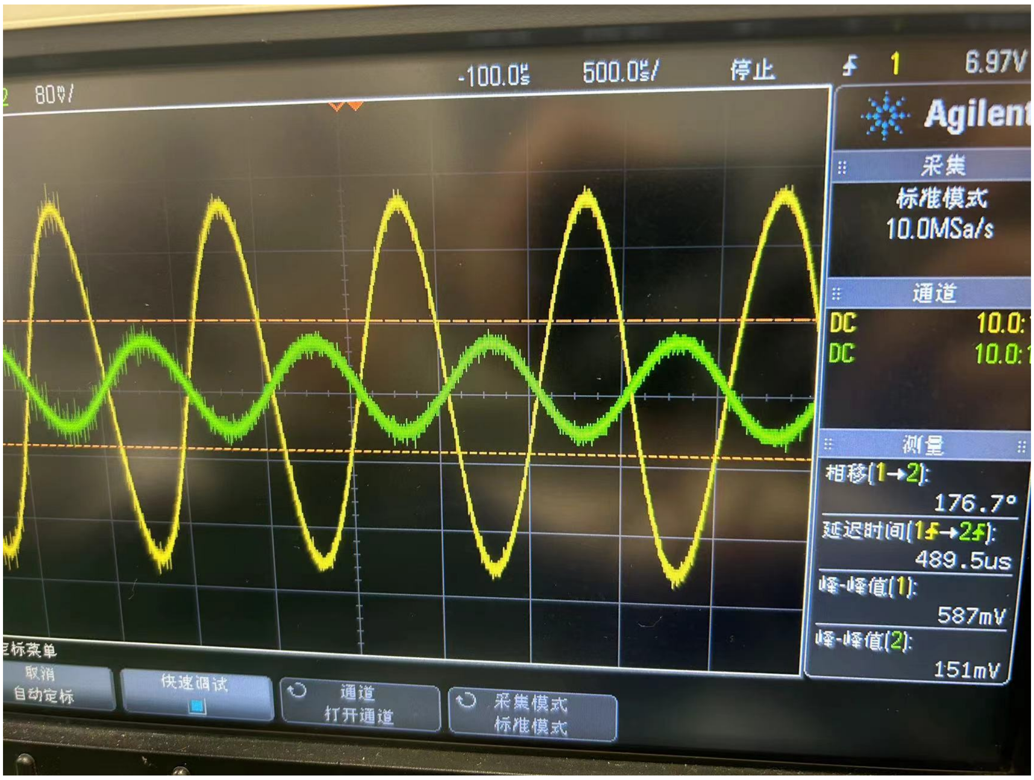

Figure 8: Vin 2

Figure 9: Vin 3

Figure 10: Vin 4

v_{in}	v_{out}	A_v
0.5	1.6	3.2
0.151	0.587	3.89
0.167	0.695	4.16
0.193	0.82	4.25

As we can see from the table, the voltage gain A_v changes with different values of the input voltage v_{in}. Initially, for $v_{in} = 0.5$ V, the gain A_v is 3.2. As the input voltage decreases, the gain increases.

2.3 Source Follower

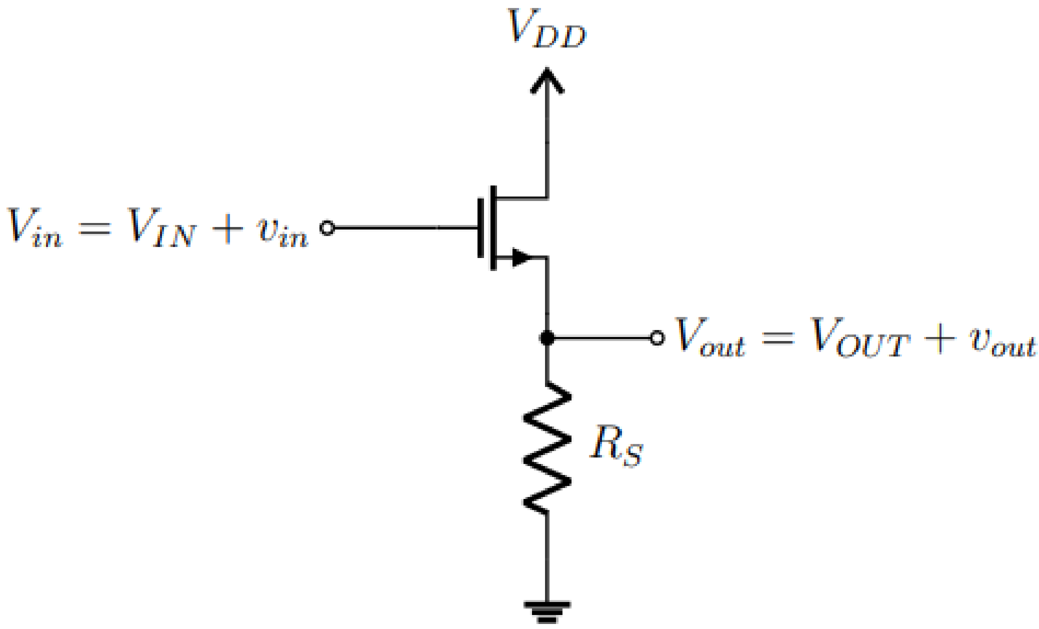

Figure 11: The Source Follower Circuit

Select Values:

I	V_{DD}	V_{IN}	R_S
1 mA	10 V	2 V	1200 Ω

Given values:

$$\begin{aligned} I_D &= 1 \text{ mA} = 0.001 \text{ A} \\ V_{DD} &= 10 \text{ V} \\ V_{IN} &= 2 \text{ V} \\ R_S &= 1200\Omega \end{aligned}$$

Using the relationships:

$$\begin{aligned} V_S &= I_D R_S \\ &= 0.001 \text{ A} \times 1200\Omega \\ &= 1.2 \text{ V} \end{aligned}$$

$$
\begin{aligned}
V_{DS} &= V_{DD} - V_S \\
&= 10\,\mathrm{V} - 1.2\,\mathrm{V} \\
&= 8.8\,\mathrm{V}
\end{aligned}
$$

$$
\begin{aligned}
V_{GS} &= V_{IN} - V_S \\
&= 2\,\mathrm{V} - 1.2\,\mathrm{V} \\
&= 0.8\,\mathrm{V}
\end{aligned}
$$

Theoretical and Actual Values

	V_S	V_{DS}	V_{GS}
Theoretical	1.2 V	8.8 V	0.8 V
Actual	0.511 V	9.53 V	1.504 V

Finally, we apply a small AC voltage on the input. The results are shown as follows:

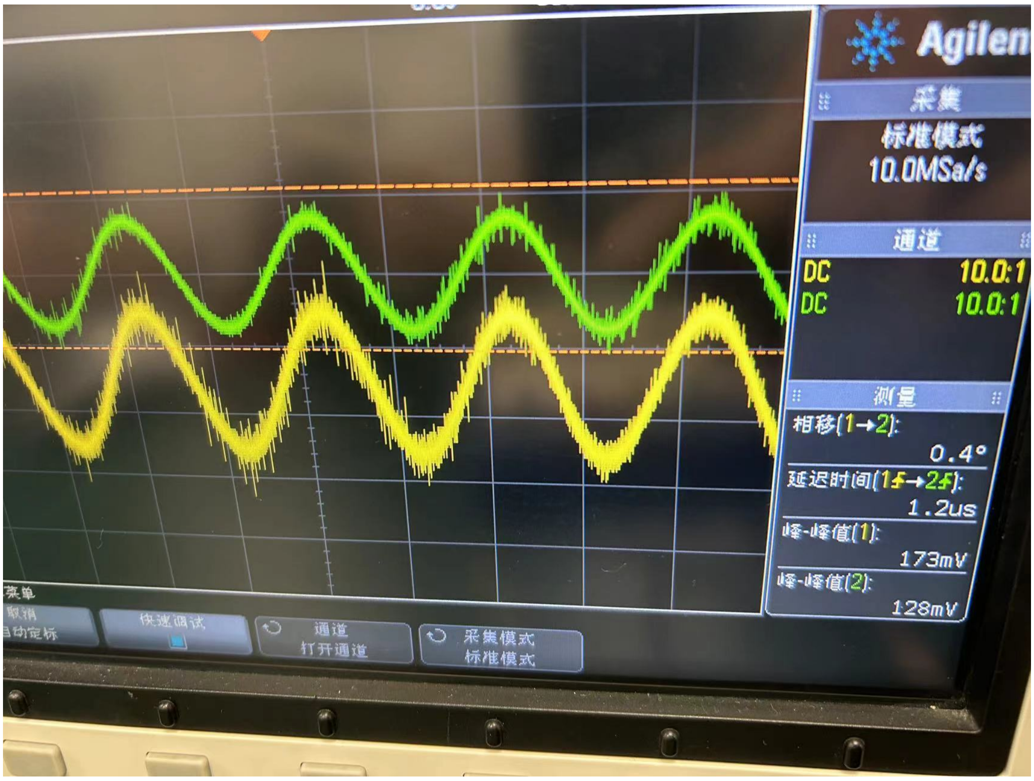

Figure 12: Vin 2-1

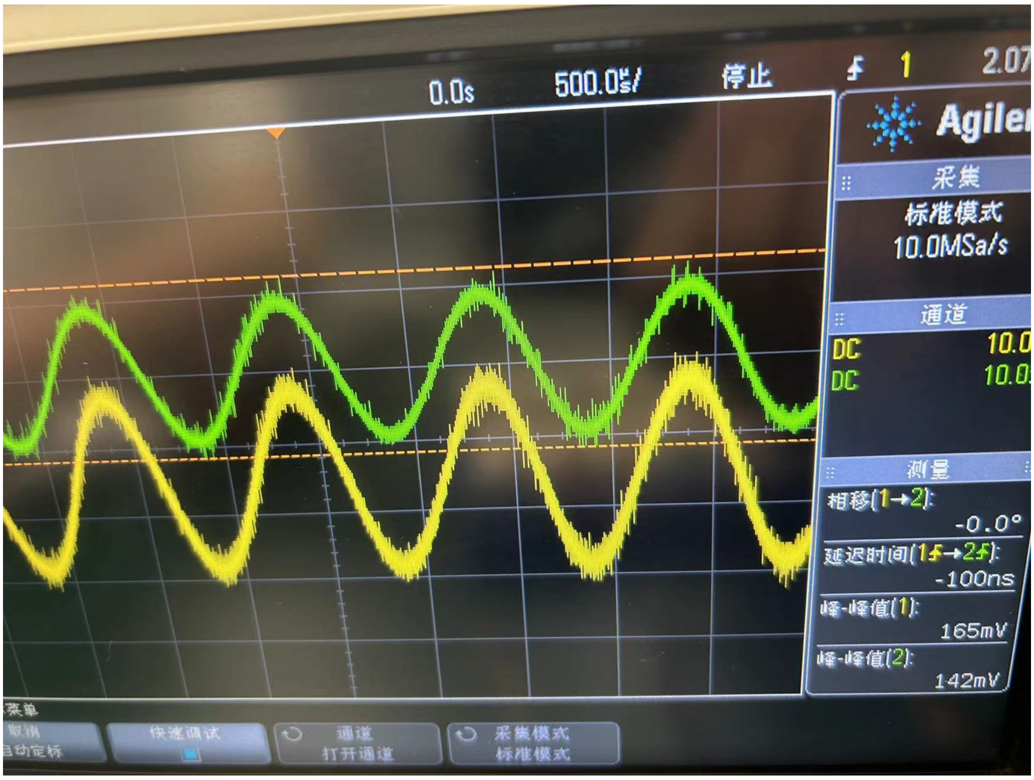

Figure 13: Vin 2-2

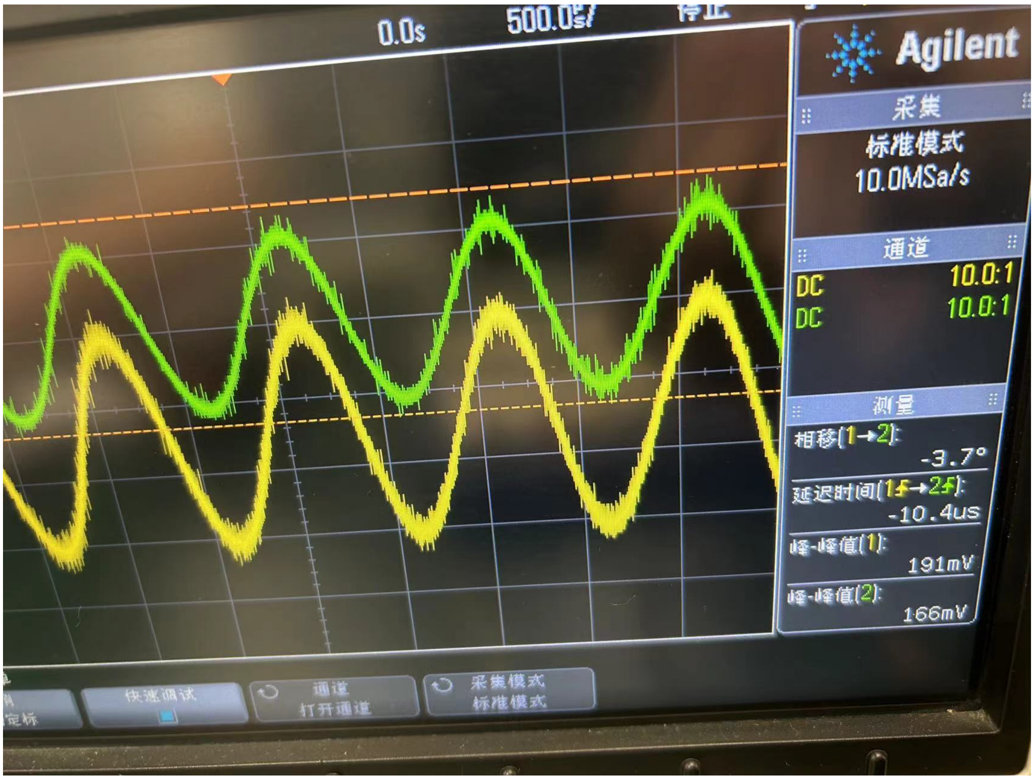

Figure 14: Vin 2-3

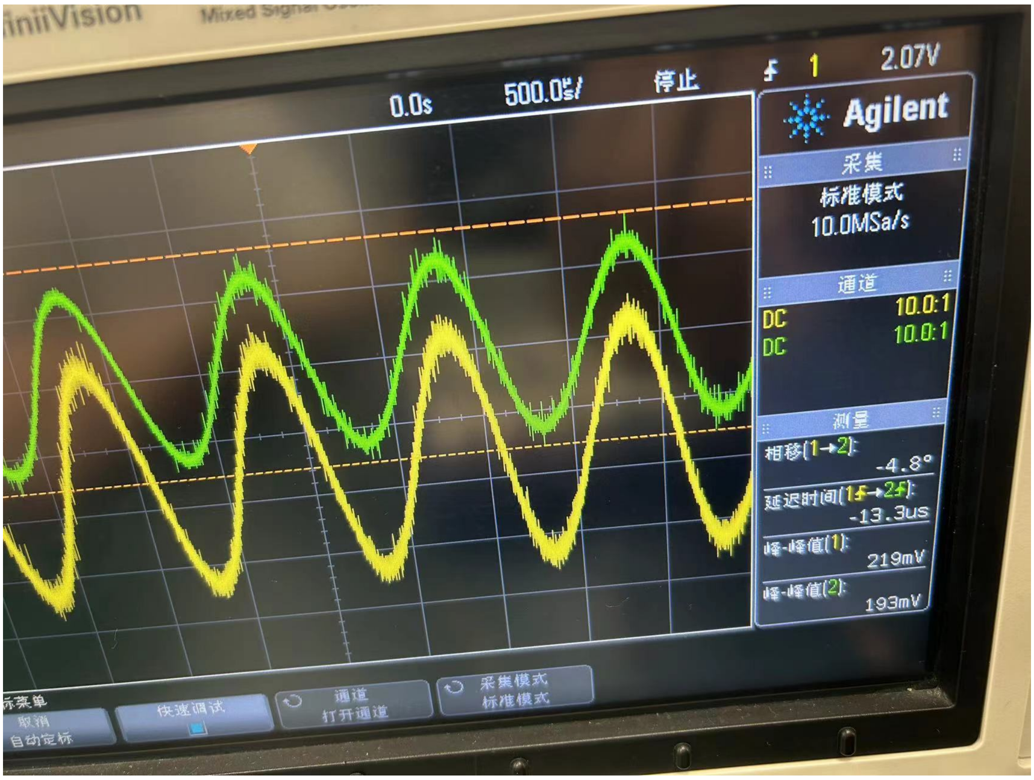

Figure 15: Vin 2-4

v_{in}	v_{out}	A_v
0.128	0.173	1.35
0.142	0.165	1.16
0.166	0.191	1.15
0.193	0.219	1.13

As we can see from the table, the voltage gain A_v changes with different values of the input voltage v_{in}. Initially, for v_{in} = 0.128 V, the gain A_v is 1.35. As the input voltage increases, the gain decreases, but it's value is still around 1.

3 Simulation Results

3.1 I-V Characteristics of 2N7000

The circuit for measurement built in the simulation is as follows:

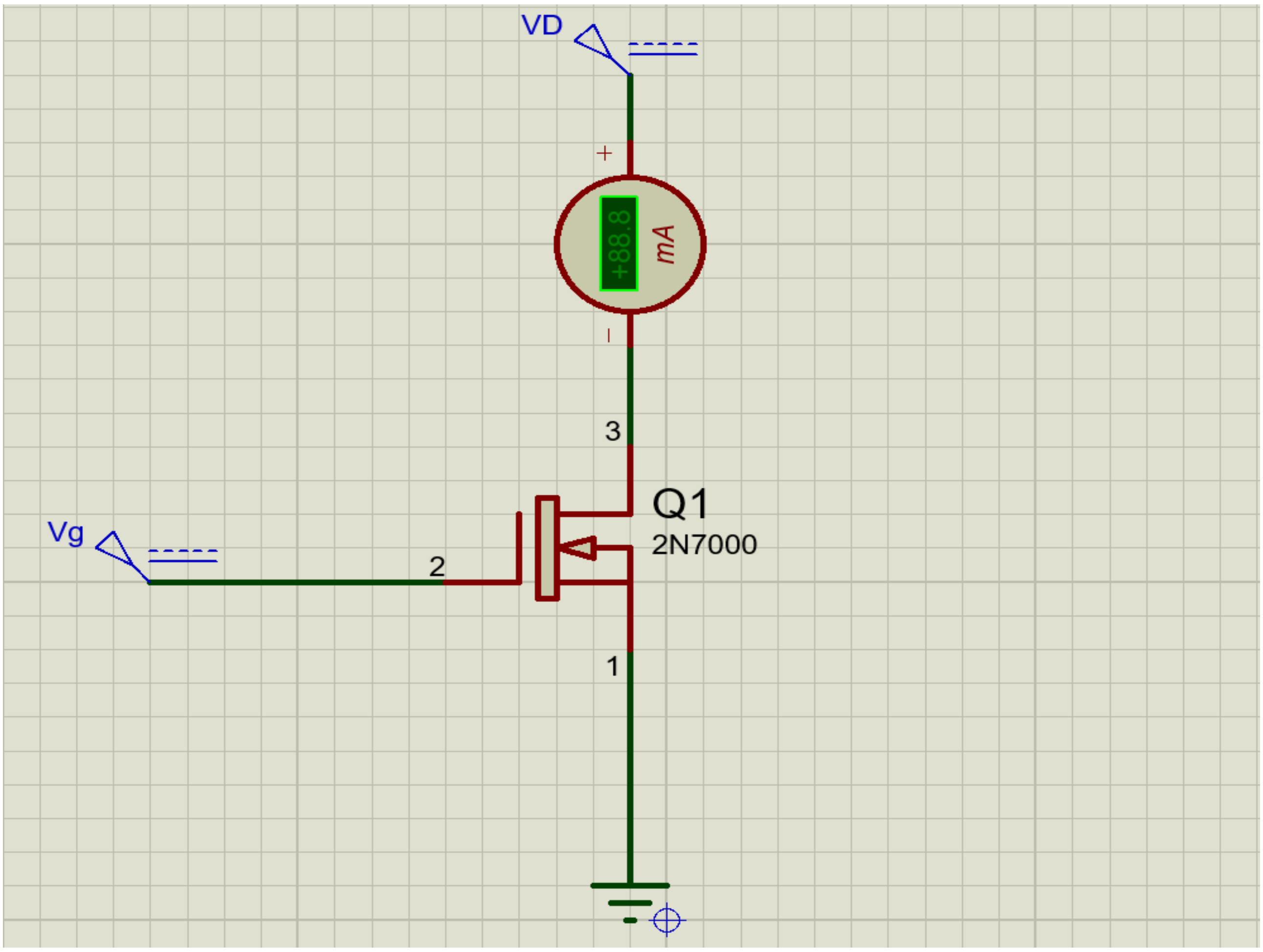

Figure 16: I-V Characteristic Testing Circuit in Simulation

Vg=1V							
Vd(V)	0	0.1	0.6	1.1	1.6	2.1	2.6
I(mA)	0	0.38	0.39	0.41	0.42	0.44	0.45
Vg=1.2V							
Vd(V)	0	0.1	0.6	1.1	1.6	2.1	2.6
I(mA)	0	2.25	3.68	3.73	3.77	3.81	3.85
Vd=1.5V							
Vg(V)	0.8	0.9	1.0	1.1	1.2	1.3	
I(mA)	0	0.02	0.42	1.67	3.76	6.67	
Vd=2V							
Vg(V)	0.8	0.9	1.0	1.1	1.2	1.3	
I(mA)	0	0.02	0.43	1.7	3.8	6.73	

Figure 17: Simulated I-V Statistics

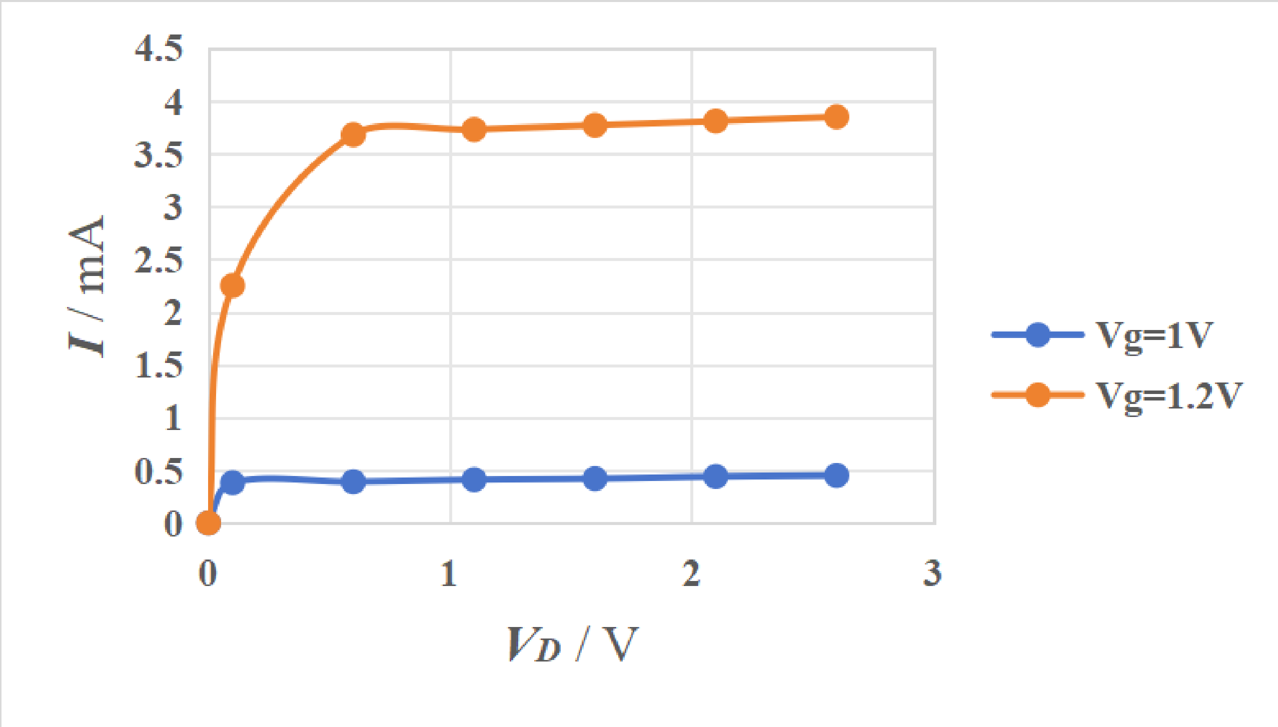

Figure 18: Simulated I-V Curve of 2N7000 (1)

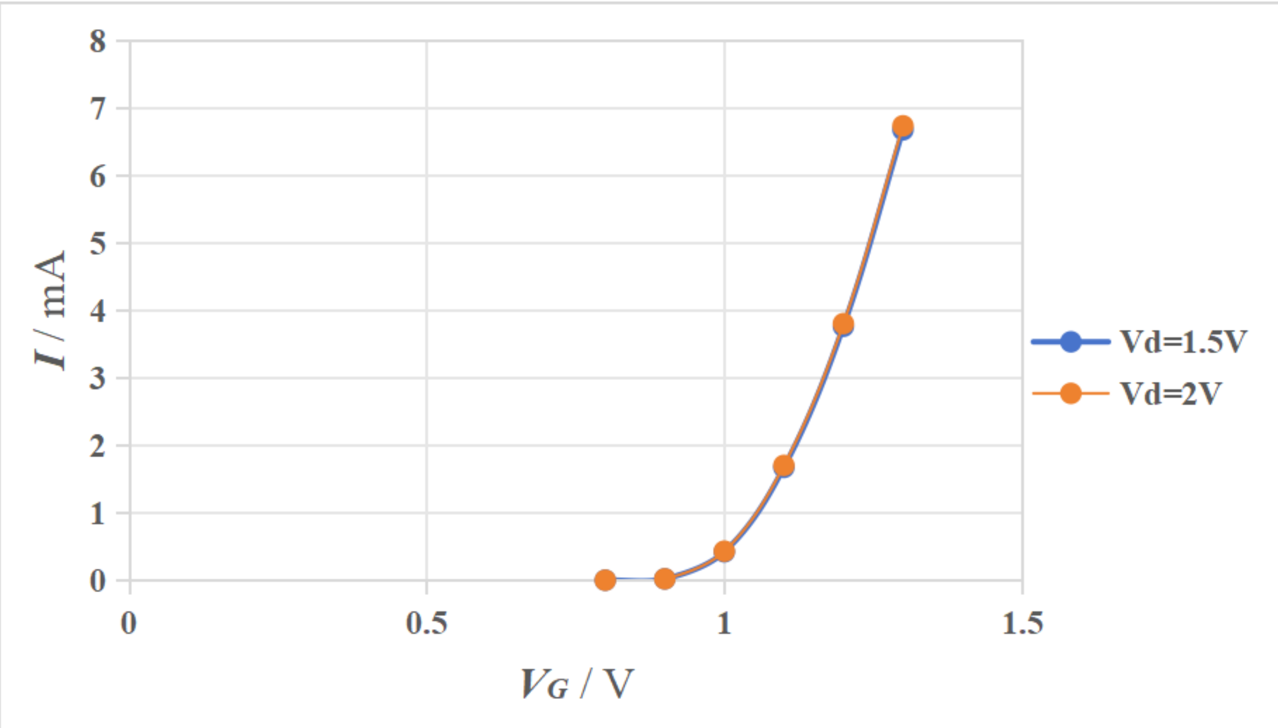

Figure 19: Simulated I-V Curve of 2N7000 (2)

3.2 Common-Source Amplifier with Source Degeneration

3.2.1

The simulation circuit to test the DC circumstance similar to the actual experiments is as follows:

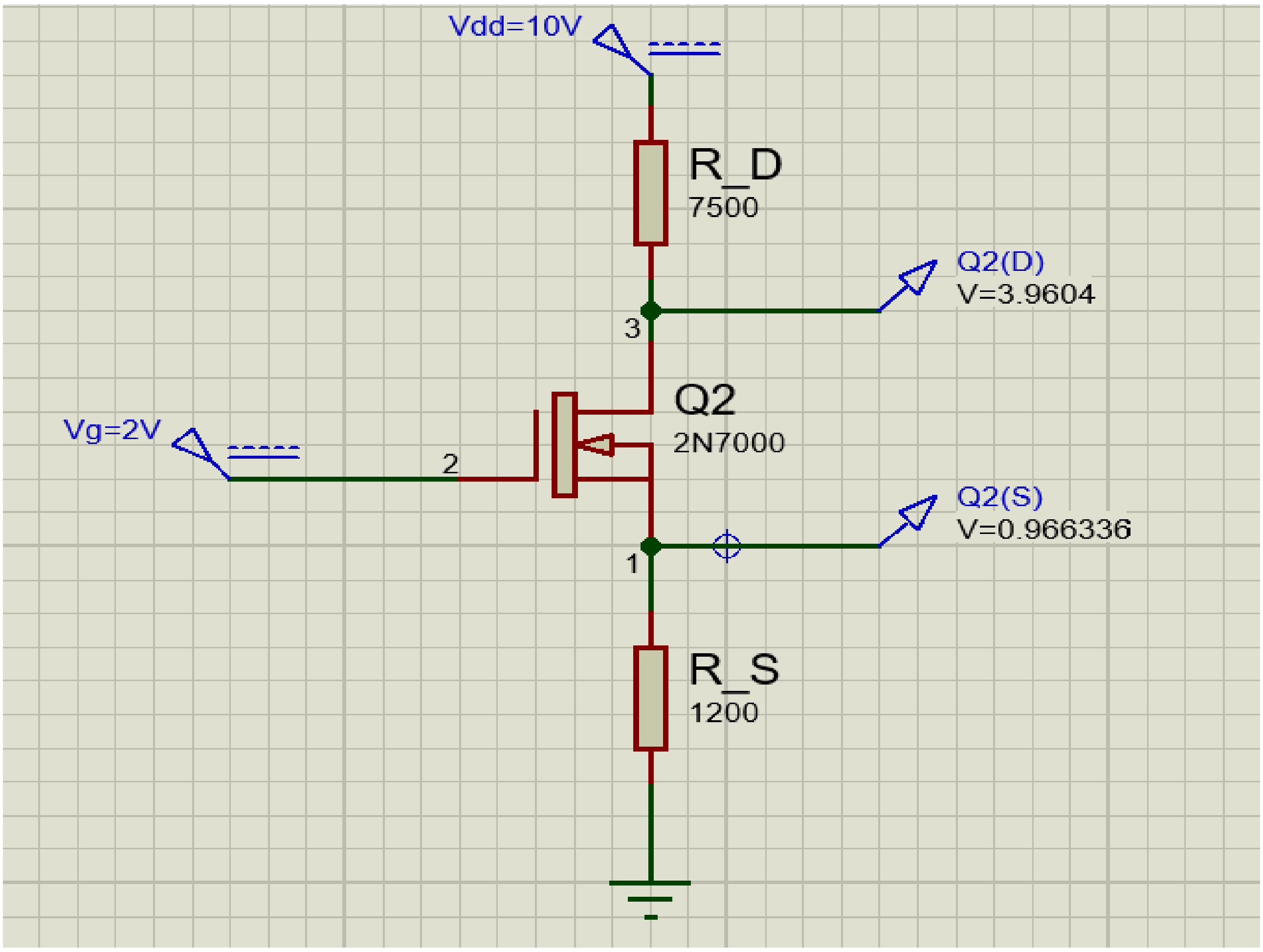

Figure 20: DC Sweep Testing Circuit

3.2.2

According to the circuit above, the obtained simulation statistics are:

	V_D (V)	V_S (V)	V_{DS} (V)	V_{GS} (V)
Theoretical	2.50	1.20	1.30	0.80
Actual	3.96	0.97	2.99	1.03

Figure 21: Measurement of the DC Characteristics

3.2.3

With the small AC signal (1kHz) added to the input terminal, the simulation circuit is modified as follows:

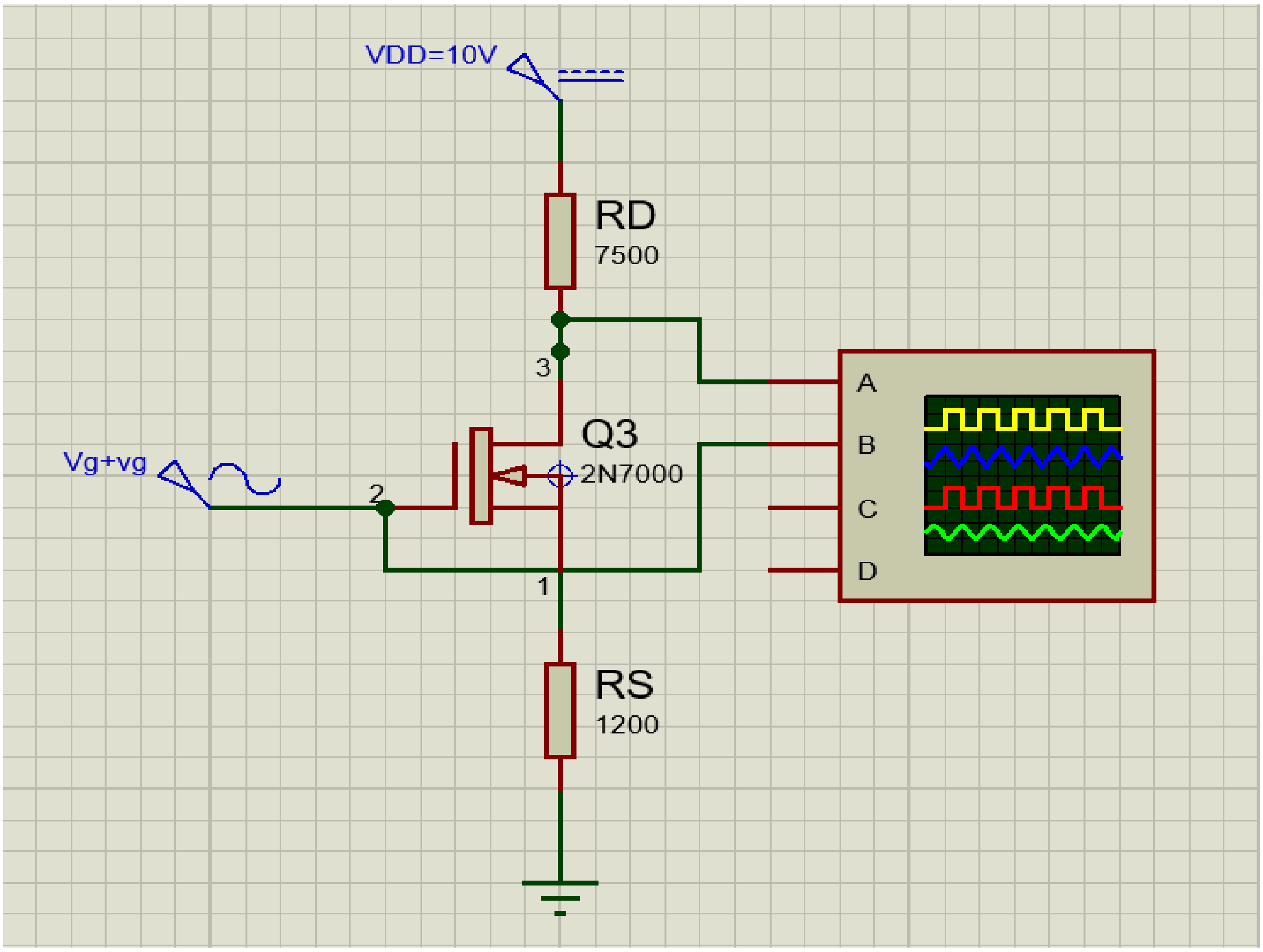

Figure 22: Simulated Common-Source Amplifier Circuit

v_{in}(V)	0.100	0.150	0.200	0.250
v_{out}(V)	-0.570	-0.855	-1.140	-1.430
A_v	-5.70	-5.70	-5.70	-5.72

Figure 23: Simulated Output Gains

3.3 Source Follower

3.3.1

By DC analysis, the simulation circuit implement the source follower is as follows:

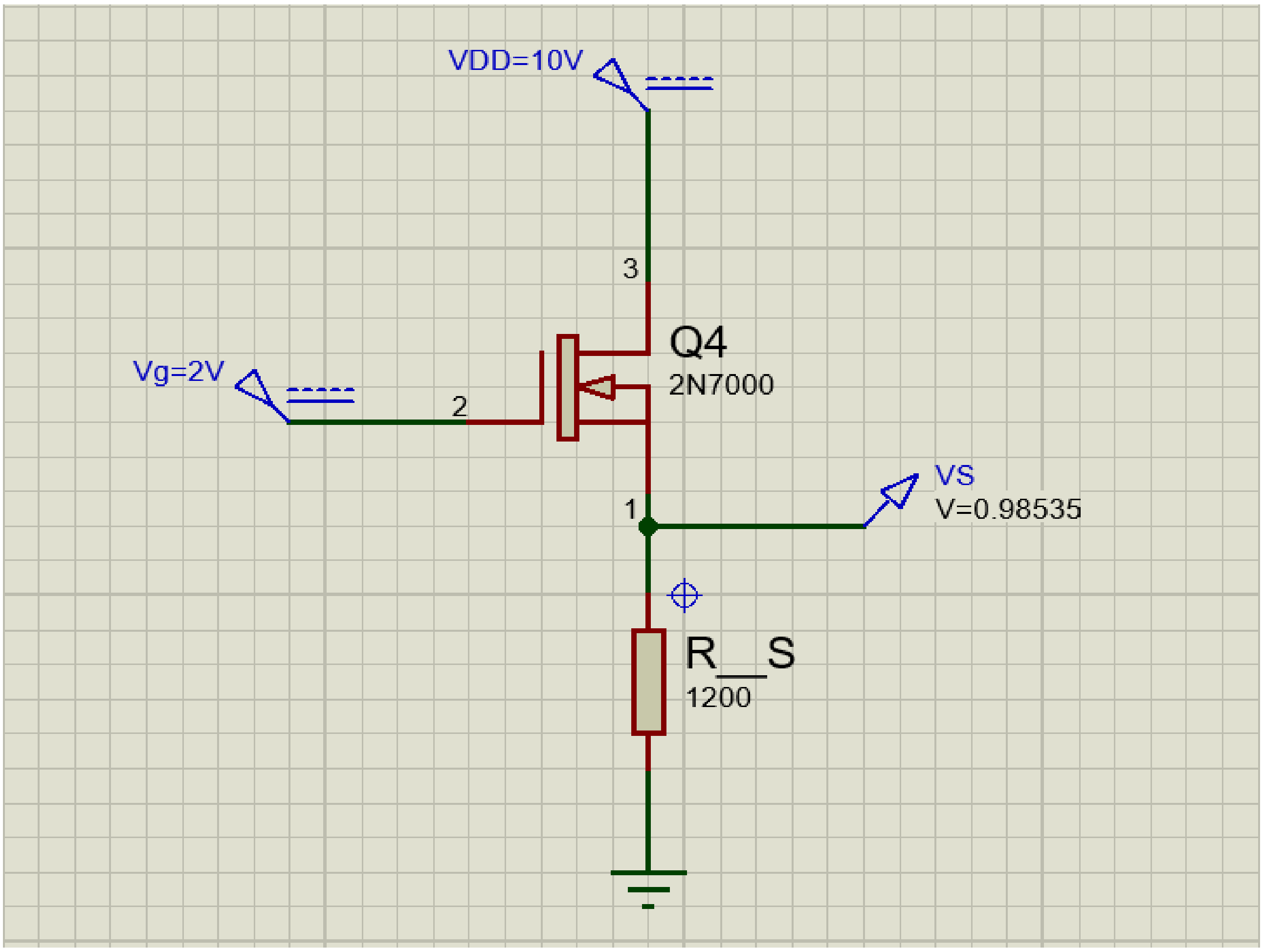

Figure 24: DC Testing Circuit of the Source Follower

3.3.2

	V_S (V)	V_{DS} (V)	V_{GS} (V)
Theoretical	1.20	8.80	0.80
Actual	0.99	9.01	1.01

Figure 25: Measurement of the Statistics in DC Sweep

3.3.3

The simulation circuit of the source follower is modified for the small AC signal (1kHz):

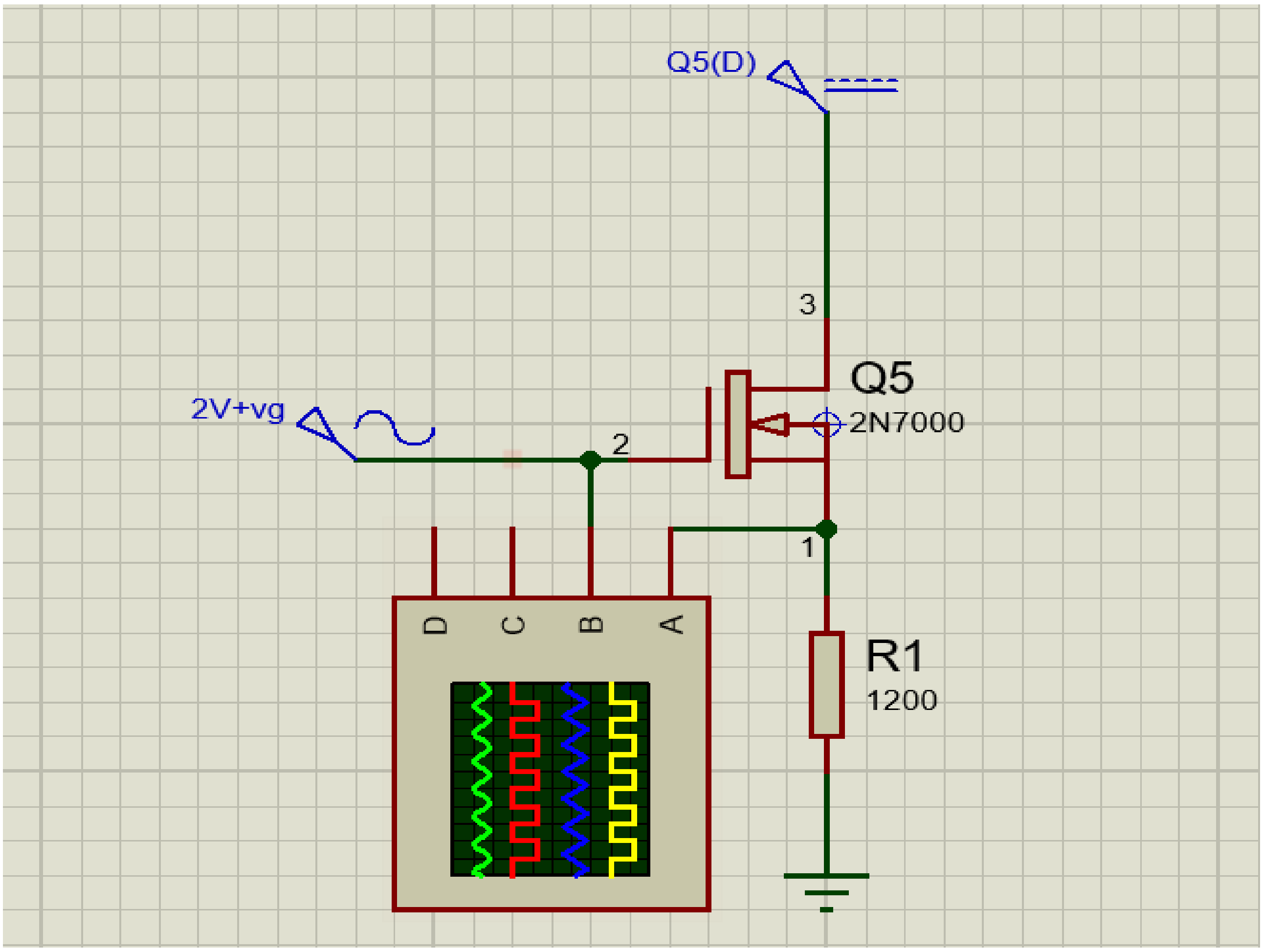

Figure 26: Simulated Source Follower Circuit

v_{in}(mV)	100	150	200	250
v_{out}(mV)	92.5	137.5	187.5	232.5
A_v	0.925	0.917	0.938	0.930

Figure 27: Simulated Output Gains

4 Error Analysis

4.1 I-V Characteristics of 2N7000

During the real experiments, the reading of the voltage meter could hardly remain stable and wandered within a relatively large range, and both the readings of the voltage meter and the current meter were influenced.

Besides, the MOSFET unit (2N7000) would reach a high temperature as if burning during the experiments, and the well-functioning circuit would sometimes suddenly turn into an open circuit that was irreparable unless part of the units were replaced. So there might existed potential errors in such behaviors.

4.2 Common-Source Amplifier with Source Degeneration Source Follower

The coefficients of the MOSFET unit (2N7000) including $\mu, C_{ox}, \frac{W}{L_{eff}}$ might differ from the statistics recorded in the data sheet and lead to chages in the actual DC current which was different from expectation.

5 Conclusions

After the experiments and the findings in this lab, we have relatively successfully observed the expected results related to the properties of the MOSFET unit, from its basic V-A characteristics to the application circuits like the common-source amplifier and the source follower.

During the experiments about the V-A characteristics of the MOSFET unit, we directly learnt about the differences in its working status between linear mode and saturation through the visualized curves.

Besides, the experiments concerning the amplifier and the source follower helped enhance our comprehension of the small signal model for MOSFET units to achieve corresponding functions with designed output-input gain.

6 Reference

[1] ECE3110J_2024SU_Lab3Manual, ECE3110J Teaching Group

[2] 2N7000_datasheet, On Semiconductor®.